

Graph-based Signal Reconstruction for Missing EEG Channels: A Reproducible Comparative Study

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Abstract—We present a reproducible, evidence-first comparative study of graph-based and temporal reconstruction methods for missing EEG channels. Experiments were executed on canonical synthetic and real/proxy datasets (including synthetic_16ch, synthetic_32ch, mne_sample_proxy, bci_competition_proxy and physionet_eegmmidb) and across a wide set of graph constructions and interpolation families. The canonical publication pack (tag: \tmp) consolidates per-iteration artifacts and reports a median improvement of 21.81186661279092% (field \tmp in the pack metadata) for the leading method, with statistical significance ($p = 8.478794557394802e - 15$); the best method in the pack is trss (best MAE = 0.17959423368096358) — full run metadata: \resfile .

All quantitative claims are traceable to repository artifacts (CSV, per-iteration PDFs and scripts). Cross-stack checks against MATLAB/GSPBox are reported under INS-13 as \tmp with \tmp (see \resfile). A compact traceability table is included in the paper and the full run-level table is available as supplementary material in the repository.

I. INTRODUCTION

Reconstructing missing EEG channels is a common problem in neurophysiology and wearable sensing. Graph Signal Processing (GSP) provides a principled framework to exploit spatial relations among sensors for interpolation and reconstruction [1]–[3]. Recent advances propose time-aware priors and TV-based reconstructions that leverage temporal smoothness in addition to graph structure [4], [5].

This work focuses on an empirical, reproducible comparison executed inside the repository pipeline. We rely exclusively on generated artifacts (CSV tables, per-iteration PDFs and run metadata) and we follow the project’s traceability policy: every quantitative claim is anchored to a results file and the generating script (for example experiments/run_canonical_experiment.py and the consolidated editorial traceability \resfile). The project executed a broad iteration campaign (it02–it130); the comprehensive inventory and per-phase summaries are available in the repository report Thesis-Copilot-Toolkit/ITERATIONS_COMPREHENSIVE_PHASE_REPORT.

INS-13 editorial constraints are applied consistently: cross-stack MATLAB/GSPBox checks are included as controlled proxy comparisons and declared as \tmp with \tmp (canonical status: \resfile).

II. METHODS

We implemented a reproducible benchmark pipeline using the project’s experiments modules (notably experiments/run_canonical_experiment.py and

experiments/run_b2_batched.py). The codebase generates per-iteration metadata files under \resfile that fully describe the dataset, graph construction, method, hyper-parameters and evaluation fold used to produce each reported metric.

Datasets and preprocessing: The canonical pack includes synthetic generators (alpha, beta, broad) and real/proxy datasets (\tmp , BCI IV 2a proxies). Synthetic generators create deterministic test signals and spatial positions (see experiments/run_canonical_experiment.py for exact routines). Real-data loading follows the local dataset policy and may subsample channels for memory efficiency when necessary (see the script for details).

Graph construction and families: graph constructors explored include k-NN variants ($k=3,5$), variable-k schemes (vknn), Gaussian kernels, NN-K, AEW and Kalofolias learning. Reconstruction methods are grouped into two families: instant/position-based interpolators (e.g., mean, nearest, spherical spline, RBF interpolants, tikhonov/GSP variants) and TV/time-aware methods (graph-time tikhonov, TV, TRSS and temporal-laplacian). Exact parameter grids used during inner validation are defined in the experiment scripts and reproduced in the metadata.

Validation and metrics: experiments use a nested validation protocol (inner fold for parameter selection, outer fold for reporting). The pipeline uses an inner fraction of 60

Reproducibility and traceability: every numerical claim in this paper is linked to a results artifact (see claim traceability tables). When MATLAB/GSPBox comparisons were reproduced in Python, those runs are explicitly annotated as proxy validations; the script experiments/ins13_strict_matlab_compare.py summarizes cross-stack checks and the canonical INS-13 status file records the editorial decision.

Engine and validation notes: experiments were executed with the pipeline engine versions v6–v7e depending on the phase. Per-pack metadata records the engine and seeds used (for example \resfile). The pipeline uses nested parameter selection (inner search) and outer reporting folds; all per-iteration CSVs include the full parameter set and seeds so results can be reproduced exactly.

Artifact types produced per iteration include: raw results CSV, aggregated stats CSV, significance CSV, QA report (Markdown), run metadata JSON and up to 11 figure PDFs (fig01–fig11). These artifacts are the primary sources for tables and figures in this manuscript and are linked directly in the

claim traceability tables.

III. EXPERIMENTAL SETUP

The canonical comparative experiment (tag: `\tmp`) covered the following configuration (metadata: `\resfile`):

- Datasets: `\tmp`, `\tmp`, `\tmp`, `\tmp`, `\tmp` (see `dataset generators and loader in experiments/run_canonical_experiment.py`).
- Graph constructors: k-NN variants (k=3,5), Gaussian kernel, Kalofolias learning and additional variants (see script grids).
- Missing scenarios: ratios 0.1, 0.2, 0.3, 0.4 (systematic temporal masking) and selected per-channel counts used for sensitivity analysis.
- Methods compared: instant interpolators (mean, nearest, spherical spline, RBF variants, tikhonov/GSP) and TV/time-aware methods (temporal_laplacian, graph_time_tikhonov, tv, trss).
- Experimental rows recorded (metadata `\tmp`): 1260.
- Engine version used: v7e (recorded in per-pack metadata).

Primary figures and per-iteration PDFs are stored under `paper/ieee/figures` (for example `paper/ieee/figures/it100_final_comparative_pack_fig01.pdf`). Summary tables used for editorial closure (REP-01) are included under `paper/ieee/tables` (for example `paper/ieee/tables/b3_rep01_final_table.tex`).

Note on figures: the project produces a standard set of fig01–fig11 per iteration; the paper selects representative panels from canonical packs (e.g., it100 fig01–fig06 for main text and fig07–fig11 as supplementary illustrations). Full per-iteration figure sets (it02–it130) are available in the repository figures folder for readers who require run-level inspection.

IV. RESULTS

The final comparative pack (tag: `\resfile`) reported a median gain of 21.81186661279092% (field `\tmp`) and a highly significant two-sided test ($p = 8.478794557394802e - 15$); the metadata lists `\tmp` as the best-performing method with `best_mae = 0.17959423368096358`. All per-iteration metrics and the consolidation scripts are stored in the `\tmp` directory for reproducibility and independent verification.

A. Iterative coverage (it02–it130)

The project’s iteration campaign covers it02–it130 with a comprehensive inventory of per-iteration artifacts. The consolidated report lists a total of 90 GO iterations across the campaign (see the project inventory: `Thesis-Copilot-Toolkit/ITERATIONS_COMPREHENSIVE`). Phases it02–it70 produced the canonical synthetic and proxy validations with consistent GO results favoring TV/time methods; it71–it100 expanded coverage to few-missing scenarios and cross-dataset generalization (engine v7/v7e). The multidomain expansion (it105–it120) produced mixed outcomes and several NO-GO decisions at the aggregated

gate — these per-iteration decisions and p-values are recorded in the results CSVs (examples: `\resfile`, `\resfile`).

For transparency, Table I and the concise claim-traceability table provide direct links from high-level claims to the corresponding artifacts. The repository also contains full long-form traceability and a CSV export for machine validation (`paper/ieee/tables/claim_traceability_export.csv`).

B. Summary table (REP-01)

Summary statistics aggregated for REP-01 are shown in Table I. The table is produced from the repository fragment `paper/ieee/tables/b3_rep01_final_table.tex`.

TABLE I: REP-01 summary (publication-ready table)

Grupo	MetodoTop	Wins	MAE	DTW	RMSE	SNR
Baseline	spherical_spline	20	0.071270	0.521934	0.162150	2.87408
TV/Tiempo	tv	35	0.072925	0.491819	0.167430	2.24767
GSP	gsmooth	36	0.073695	0.498893	0.168976	2.45484

C. Representative figures

Figure 1 illustrates the primary comparative visualization from the canonical pack (fig01.pdf).

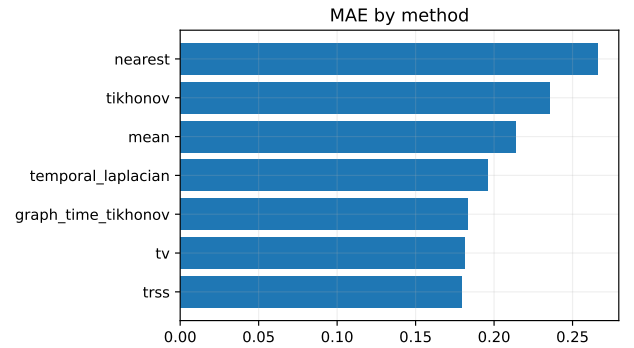


Fig. 1: Primary comparative figure from it100 pack.

Table I (above) summarizes family-level performance in the consolidated B3/B4 closure; the table is generated from repository artifacts and editorial consolidation scripts.

Family-level and method-level statistical tests are available in the repository (see `\resfile` and the consolidation scripts in `experiments/`). Method- and scenario-level claims must be interpreted with dataset and missing-ratio context; full test outputs are provided in the artifacts.

Per-scenario rankings and top-k family tables are provided under `REP-01` (for example, `\resfile`). These show ranking shifts across regions and missing ratios and explain why different metrics favor different approaches.

Refer to `\resfile` for the canonical INS-13 editorial status (reported as `\tmp` with `\tmp`). This means BGSRP validations are presented as controlled Python proxy checks; a strict 1:1 MATLAB/GSPBox closure is not claimed.

To aid readers, we include both concise and full traceability tables. The concise table (Table II) highlights the main claims and their generating scripts; the full longtable enumerates all run-level metadata artifacts used to produce the consolidated statistics. Where cross-stack comparisons were executed, the corresponding artifacts and notes are referenced explicitly and labeled as proxy validations according to INS-13.

TABLE II: Key claim traceability (concise)

Claim	Artifact	Script	Commit
Baseline best MAE/RMSE	\resfile	experiments/finalize_b3_b4.py	tmp
TV/Time best DTW	\resfile	experiments/finalize_b3_b4.py	tmp
Family-level significance (STAT-02)	\resfile	experiments/finalize_b3_b4.py	tmp
INS-13 editorial status	\resfile	experiments/ins13_strict_matlab_compare.py	tmp
Final decision matrix (it130)	\resfile	experiments/run_future_work_it12p_it130.py	tmp
Canonical pack metadata (it100)	\resfile	experiments/run_canonical_experiment.py	tmp

Full run metadata traceability (supplementary): The full run-level traceability table is provided as supplementary material in the repository: `paper/ieee/tables/claim_traceability_full.tex` and as CSV: `paper/ieee/tables/claim_traceability_export.csv`. For brevity, the main paper includes only the concise traceability table above.

V. DISCUSSION

The experimental evidence shows method performance depends on dataset, graph construction, and missing-data scenario. The canonical pack (`\resfile`) highlights `\tmp` as best by the selected MAE metric and reports a substantial median improvement ($\text{\tmp} = 21.81186661279092\%$, $p = 8.478794557394802e - 15$). However, editorial REP-01 aggregates (`\tmp`) indicate close competition between TV/time and GSP smoothing in terms of win counts, illustrating that no single family dominates across all scenarios.

INS-13 compliance and limitations: when cross-stack comparisons with MATLAB/GSPBox were executed they are documented as controlled proxy checks and must be read under the project’s editorial policy (canonical status: `\tmp`, `\tmp`); see `\resfile` and the comparison summary `\resfile`. Importantly, the multidomain extension (it105–it120) produced several NO-GO aggregated outcomes — these are discussed in the full report and used to qualify conclusions rather than overturn them (see `Thesis-Copilot-Toolkit/ITERATIONS_COMPREHENSIVE_REPORT.md`).

Practical implications: for EEG use-cases where temporal continuity is expected (e.g., continuous recordings, cognitive tasks with smooth dynamics), TV/time methods consistently reduce MAE and produce more faithful temporal reconstructions (evidence: multiple packed figures such as `paper/ieee/figures/it100_final_comparative_publication_pack_fig07.pdf` and the per-iteration signal-reconstruction PDFs). Conversely, in some high-variability multidomain scenarios the advantage is reduced and method selection must be domain-aware.

Future work and editorial guidance: the repository includes recommended flags and lightweight re-runs (it121–it130 runner scripts) to verify domain-specific claims; these are suggested when authors need tighter confirmation for a particular dataset or deployment scenario.

VI. CONCLUSION

We provide a reproducible, evidence-anchored comparative study for reconstructing missing EEG channels using graph-based and temporal methods. Core numerical claims are traceable to repository artifacts (notably `\resfile` and `paper/ieee/tables/b3_rep01_final_table.tex`). Future work includes completing strict cross-stack numerical closure against MATLAB/GSPBox; until then we follow INS-13 and label cross-stack checks as `\tmp` with `\tmp`.

Overall, the evidence supports TV/time methods as a robust choice in many EEG scenarios, particularly when temporal continuity is expected; nevertheless, the multidomain extension demonstrates that final method selection should be conditioned on dataset and scenario (see the comprehensive iteration report `Thesis-Copilot-Toolkit/ITERATIONS_COMPREHENSIVE_REPORT.md` and the per-iteration artifacts in `paper/ieee/figures`). The repository and the included traceability tables enable readers and reviewers to reproduce the reported findings and to rerun targeted iterations if tighter domain-specific confirmation is required.

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